

Making Usability Test Data Actionable! A Quantitative Test-Driven Prototyping Approach

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ABSTRACT

In recent years, the availability and use of prototyping and usability testing tools has increased massively in practice. In parallel, research introduced multiple novel approaches and tools to assist novice and expert designers in prototyping. However, existing research has not looked into how to assist designers in conducting usability tests and making the collected quantitative data directly available in prototyping tools. In this paper, we distill three design goals from literature and propose a novel quantitative test-driven prototyping approach. The approach in its current form consists of a process and two user interface artifacts supporting selected actions of our process. We conducted a formative study to gather feedback from potential users on our approach. We use the collected feedback to refine our approach, identify trade-offs, and propose future research directions. We contribute by providing a novel quantitative test-driving prototyping approach that tightly connects prototyping with asynchronous usability testing.

CCS CONCEPTS

• **Human-centered computing** → *Usability testing; Empirical studies in HCI; User interface toolkits.*

KEYWORDS

Prototyping, Usability Testing, Data-Driven, GUI

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1 INTRODUCTION

It is well-known that involving users early in software development is a significant success factor [25, 27]. In particular, an important human-centered design activity is to develop prototypes of the graphical user interfaces (GUIs) before more resource-intensive development steps are carried out (e.g., [42]). Prototyping GUIs has been a subject for academia for more than 30 years and many commercial tools exist today in practice (e.g., [4, 6, 12, 39]). In the

last years, commercial prototyping tools have attracted a growing number and served a much broader group of users beyond experienced designers, such as software developers or product managers. The growing commercial importance of prototyping is also demonstrated by the recently announced merger agreement [2] of Adobe Inc. to acquire the prototyping tool Figma [12]. Major reason for this acquisition were Figma's millions of users and the growing ecosystem established around Figma [2].

Users of prototyping tools today are more heterogeneous with different levels of design skills and experiences. For example, Lee et al. [21] report on the difficulties that novice designers face during developing GUI prototypes. Based on these difficulties, Lee et al. propose a real-time assistance system integrated into a prototyping tool to help novice designers in developing prototypes. Besides this example, there is a whole stream of research that investigates different forms of assistance systems to help novice but also experienced designers in prototyping (e.g., [7, 32]).

It is essential for successful prototype development to continuously involve users (e.g., [15]). There are several approaches to do this, with usability testing focusing on providing quantitative data being one of the most prominent. In literature, usability tests are often classified according to whether they are conducted in-person or remotely and whether there is an additional time offset. Depending on the separation in time, the terms synchronous or asynchronous usability tests (e.g., [17]) have been coined. Asynchronous remote usability tests have the advantage that they can be less resource-demanding than in-person usability tests [5, 8]. Commercial tools (e.g., [26, 40, 41]) enabling remote asynchronous testing have also been introduced and their use has increased massively in practice in the recent years. Recent studies report that practitioners also request more assistance during creating and conducting usability tests [24]. However, to the best of our knowledge existing research has not looked into the challenge of how to assist designers within prototyping tools to conduct quantitative asynchronous usability tests and make the collected data actionable in the prototyping tool. Specifically, *integration among tools* is known to be a key driver for tool adoption [34]. In order to address this research gap we explore the question of *how assistance for designers to perform usability testing can be closely integrated into prototyping tools and resulting usability test data can be made actionable for prototyping*.

We contribute by proposing a novel approach to assist designers within prototyping tools to conduct quantitative asynchronous usability tests and provide actionable support for prototyping on the basis of the collected data and component annotations. Our approach not only simplifies the collection of high quality usability test data during prototyping by supporting annotation of components and defining prototype-centric tasks. In addition, it directly

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connects the testing results with the prototype and its major components. To come up with our approach, we distilled three design goals from literature, created a technological probe in the form of a process and two artifacts, and conducted a formative study to collect feedback from potential users. We use the collected feedback for refining our approach and describe design trade-offs. We believe our approach as a basis for stimulating discussion at the intersection of prototyping and usability testing and provides solid groundwork for future research.

2 RELATED WORK

2.1 Usability Testing

Conceptualization of Usability Testing. Usability testing of prototypes has been extensively covered in the academic literature (e.g., [15, 16, 30]) and is supported by a broad spectrum of commercial tools (e.g., [23, 26, 40, 41]). In academic literature, usability testing often is distinguished along separation of time and location between test facilitator (e.g., the prototype designers) and the test evaluator (i.e., test users or test subjects). *Remote usability testing* introduced a separation in places between test facilitators and users. In academic literature, remote usability testing started to receive more attention beginning in the mid-1990s (e.g., [16, 17]). A key benefit of separation in places is the demand of fewer resources compared to in-person usability tests (e.g., [10]). Using the term *asynchronous remote evaluation*, Hartson et al. [17, p. 229] proposed the idea of having the test facilitators and users evaluating a system being separated by space and time. Remote testing not separated in time is called synchronous remote usability testing. Previous studies found asynchronous usability testing often the least resource demanding, when compared to in-person usability tests, or synchronous usability tests [5, 8].

Task Definition in Usability Testing. A lot of research has been conducted on how usability tasks should be formulated, assigned and prioritized for evaluators to support discovering relevant usability issues. Bruun et al. compared giving evaluators predefined tasks vs. self-driven exploration without predefined tasks [9]. In their study, evaluators completing predefined tasks found more usability problems when compared to evaluators not following predefined tasks. Følstad et al. [13] argue that designers generating usability tasks need work-domain knowledge. They further suggest that certain methods (i.e., stakeholder analysis, context-of-use analysis, surveying existing users, user field observations, diary keeping, tasks analysis) should be used for the "elicitation of work-domain knowledge" [13, p. 2020]. For task assignment, Hornbæk et al. [18] find that presenting evaluators with the task of considering the application's business goals can lead to evaluators detecting usability problems that have a higher relevance to the success of the application. Researching how usability tasks should be prioritized, Abrantes et al. [1] recently proposed working along two main dimensions, *criticality*, and *relevance*. Before conducting usability tests, characteristic tasks should be created for each functionality, which is rated according to criticality [1, p. 67]. For *relevance* Abrantes et al. propose an *identified issue's impact value* ranging from 1 (suggestion) to 4 (blockage), [1, p. 70]. Abrantes et al. state they base their *identified issue's impact value* on the *impact score*

by Sauro et al. [38], the *severity of usability problems rate scale* by Nielsen [30, 31], and the *problem severity rating* by Rubin et al. [36].

2.2 Prototyping Assistance

Methods and tools to support prototyping GUIs are repeatedly subjects in academia. In parallel, a plethora of tools have emerged that support prototyping GUIs based on the demand from practice. While one might argue that even general-purpose tools (e.g., MS PowerPoint [28] or Adobe Illustrator [3]) could be considered tools that support prototyping GUIs, we will focus on dedicated prototyping tools for the remaining paper. Dedicated prototyping tools in practice usually support different levels of fidelity of the prototypes designed. There are popular examples of low-fidelity prototyping tools (e.g., [6]) and multi-fidelity prototyping tools (e.g., [4, 12]).

In addition, there have been multiple research efforts to provide assistance for designers supporting the usage of prototyping tools. Contemporary commercial tools such as Figma [12] provide the possibility to develop plugins for dedicated assistance functions. Lee et al. sort approaches, that assist GUI prototyping into three categories: First, approaches that help the users to browse examples. Second, approaches that allow the transfer from components from examples in the prototyping environment, and third, approaches that automate selected design tasks during prototyping [21, p. 2]. Lee et al. present a browser-based prototyping assistance tool aimed at supporting novice designers during prototyping GUIs called *GUIComp* [21]. *GUIComp* offers real-time feedback that consists of an attention map, suggestions of similar GUIs from RICO [11], and seven automatically computed metrics (i.e., element balance, alignment, color unity, font size and type, element size, and density). Bertram et al. [7] present a concept for the automated evaluation of GUI prototypes using metrics and a model-based evaluation of actions, tasks and scenarios predicting the usage time. They tested the concept by building a plugin for Figma.

3 A QUANTITATIVE TEST-DRIVEN PROTOTYPING APPROACH

3.1 Design Goals

To address the gaps identified in existing work, we distilled the following three design goals (DGs) for a quantitative test-driven prototyping approach:

Design Goal 1: Assist designers in annotating components (DG1-A) and defining tasks for usability testing (DG1-B) tightly integrated in the prototyping tool (DG1-C).

We decompose this design goal in three sub-design goals: Following [1], we propose that designers should explicitly annotate components (**DG1-A**). While annotating (at least the components that are required for key interactions of evaluators) introduces additional work for the designers, annotations make the analysis of logged data more accessible. Automatic analysis of, e.g., mouse movements over a component may be improved if the target component is annotated, making its desired use known for automated analysis. In order to reduce the effort of manual annotation, approaches to automate annotations such as [22] have been proposed and may be leveraged.

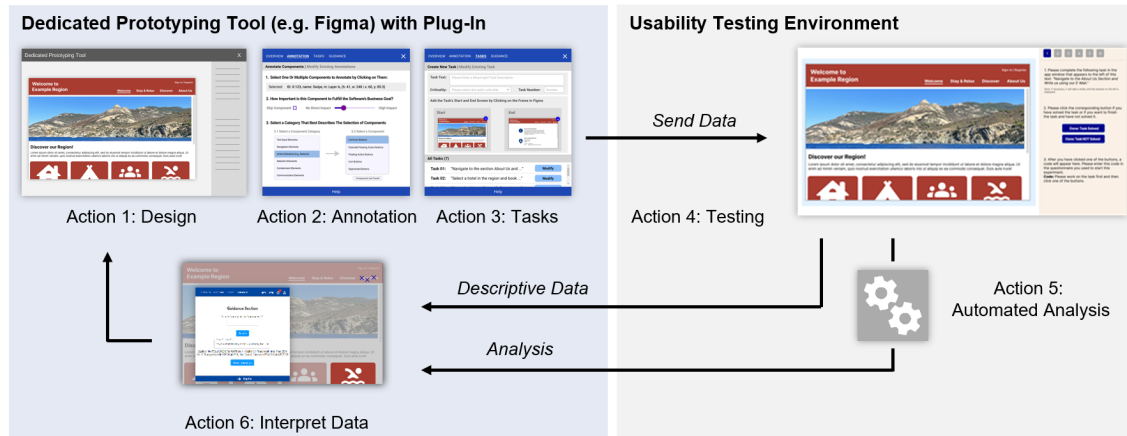


Figure 1: Proposed approach for integrating component annotations, tasks, asynchronous usability tests, and test results into prototyping tools. Designers use prototyping tools to build their GUI prototypes (*Action 1*). Using a plugin in the prototyping tool, they can annotate components (*Action 2*) and generate tasks for usability testing (*Action 3*). The prototype is tested asynchronously and remotely (*Action 4*). Descriptive data and data from further analysis steps (*Action 5*) are sent back to the prototyping tool where actionable support for designers to interpret the results from tests is provided (*Action 6*).

Bruun et al. [9] have shown that task assignments and instruction types influence the number of usability problems identified by evaluators in asynchronous usability testing. This is also enforced by usability testing tools from practice (e.g., [23, 26]). Thus, we propose that designers should be supported in defining predefined tasks within the prototype tool to prevent that users explore the prototype on their own independent of defined tasks (**DG1-B**).

Palani et al. [34] report *integration with existing workflow* as, on average second out of seven dimensions of a practitioners' value framework and adoption criteria for creativity support tools. In their questionnaire-based study participants ranked *integration with existing workflows* before e.g., *performance*, *usability*, *level of support*, and *financial cost* [34]. Within *integration with existing workflows*, Palani et al. report that *integration across tools* was emphasized as most important by participants. Integration across tools can be achieved "[...] through plugins, exporting and importing features, etc." [34, 6th page]. We therefore propose that annotating components and defining usability tasks should be integrated in the prototyping tool (**DG1-C**).

Design Goal 2: Support designers in collecting quantitative usability test data asynchronously for the prototype based on the defined tasks.

To follow a usability test-driven approach for improving the prototype, there is a need to support performing usability testing and collect corresponding data. Generally speaking, various forms of usability testing could be applied, and collected data fed back into the dedicated prototyping environment. We focus our approach on quantitative forms of usability testing and suggest asynchronous usability tests comparable to approaches in literature (e.g., [35, 43]).

Design Goal 3: Present the collected data from usability tests in the prototyping tool (DG3-A) directly connected to the annotated components (DG3-B) in a specific, justified and actionable form (DG3-C).

We decompose this design goal into three sub-design goals as follows: First, the usability tests' results should also be tightly integrated into the workflow of the prototyping tool [34] (**DG3-A**). Second, the collected test data should be connected to the prototype components in the prototyping tool (**DG3-B**). Thereby, designers are empowered to evaluate whether individual components were used the way they were intended and designers are guided to first focus on components they previously identified as critical to the applications business goal. Finally, the data collected during tests should be presented in an effective and possibly even interpreted form that enables designers to improve their designs. Comparable to effective qualitative feedback which "in general, [...] is specific, actionable, and justified" [29, p. 55], the data presentation and interpretation ideally would aim for the same qualities (**DG3-C**).

3.2 Quantitative Test-Driven Prototyping Process Model

Building on our our design goals, we propose a test-driven prototyping process model that defines key actions to assist designers and integrates usability testing into prototyping tools. Our process model is shown in Figure 1 and comprises six actions. Those actions can be performed in the presented order, but since prototyping includes revisiting steps during design, and especially when designers receive feedback, our proposed actions are designed to be repeated in an iterative fashion. Deviations from the defined order proposed are possible. We describe the actions as follows:

Action 1: Designers generate the first iteration of their prototype using a dedicated prototyping tool. For the consequent actions, the prototyping tool used by the designers must allow third-party software to interact with the design via e.g., plugins (**DG1-C**).

Action 2: Following design goals DG1-A and DG1-C, designers annotate components directly in the dedicated prototyping tool. They thereby reflect on the importance of each component they select to annotate and provide a basis for subsequent (automated)

analysis of the data from usability testing. Since the use of different components (e.g. *Navigation Bar Item* and a *Slider*) will result in different usage patterns, *Action 6* can profit on a component level if the components' design intents are known. The annotation may be supported by automated approaches such as [22].

Action 3: Following DG1-B, DG1-C and thereby laying the ground for DG2, in this action, designers define tasks for the usability tests with regards to the designed prototype and its components.

Action 4: To serve DG2, the prototype with the annotations and tasks has to be exported to a usability testing tool. Different forms of usability tests can be applied. In our process, we propose an automated export via application programming interfaces to usability testing tools that implement the asynchronous remote usability tests and allow for the vice-versa return of the (processed) results to the prototyping tool. Such usability test tools can be realized as web apps that present the design prototype as a clickable website, show the design task and collect interaction data. Recording usability tests on websites and interpreting logged usage data were described in literature more than 20 years ago (e.g., [35, 43]). We suggest to seamlessly integrate these well established usability testing processes and tools into prototyping tools.

Action 5: The quantitative interaction test data collected in the previous action is analyzed during this action following DG3-C. Since interaction data is connected with annotated design components, this action aims at automatically generating recommendations for usability issues of selected components. While we have not yet developed a running prototype for this action, related work (e.g., [20]) makes us confident that automatic usability issues detection based on interaction data for selected components can be achieved.

Action 6: Raw data from usability tests is sent back to the prototyping tool and presented in an actionable way (DG3-A, DG3-B). Interaction data can be linked directly to components and presented in an aggregated form. Additionally, the success of the task performance can be visualized following approaches such as [1, p. 71] to support designers in prioritizing usability issues. Furthermore, with *Action 5*, data can be presented and, to a limited extent, interpreted to achieve DG3-C better.

3.3 Quantitative Test-Driven Prototyping Artifacts

To instantiate our overall approach, we designed multiple artifact prototypes in the form of users interfaces that we demonstrated in combination with the approach to our participants in the formative study. Figure 1 depicts how the components interact. In the following, we explain the user interfaces that designers interact with to annotate components and set up remote usability tests in the prototyping tool using our approach (*Action 2* and 3). The current version of both interfaces is shown in Figure 2. Both user interfaces were designed to instantiate DG1 so that they can be integrated as a plugin into prototyping tools. In our case we rely on Figma.

Artifact supporting Action 2 (Annotation). Designers can select one or multiple components in Figma and check their selection (A in Figure 2). A functional component can be composed of several individual elements in Figma (e.g., a button can be composed of a rectangle and a text). Next designer assess the component's importance to the application's business goal in the context of the current

prototype screen (B in Figure 2). Ultimately designers complete annotating the functional component by selecting a category that best describes that component (C in Figure 2). For our prototype interface we included predefined component categories. To comply with standards from practice we based our categories on those in Google's *Material Design* [14].

Artifact supporting Action 3 (Task Design). Designers can use this interface to create new tasks and modify existing ones. When creating a task, the first step (D in Figure 2) is to define a task as text. Additionally, the designer evaluates the criticality of the task for the business goal of the application. For this, we have adopted the suggestion of Abrantes et al. and chosen the categories: "not a critical function", "nice to have functionality", "necessary task, but there are different ways of accomplishing it", "absolutely necessary to accomplish the prototype's goals" [1, p. 67]. Subsequently, the designers mark the first screen of the task and the screen reached when the task is completed (E in Figure 2).

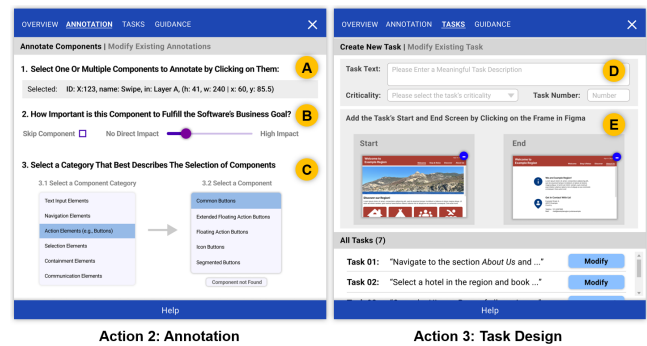


Figure 2: Proposed user interface for our assistance plugin for the prototyping tool Figma supporting the actions: Annotation and Task Design. For *Annotation*, designers select one or multiple components in Figma, check their selection (A) and start annotating: (B) the component's criticality for the application's business goal, (C) the component's category and type. For *Task Design*, a task relevant to the application's business goal is textually described (D), and the task's criticality is rated. Finally, the designer selects the first and last screen (E) in Figma that must be reached to solve the task.

4 EVALUATION STUDY

4.1 Formative Study of Process Model and Artifacts

Set-Up. We conducted a formative study to evaluate the proposed process model and the artifacts. We used the literature-grounded initial version of the process and associated artifacts as a technological probe [19]. The technological probe was used to collect participants' reflections, requirements, and feedback. In line with Ruoff et al. [37], our interview strategy specifically focused on identifying design and requirement trade-offs. In our formative study, we let participants interact with and reflect on our technical probe (i.e., the artifacts), but also by reflecting on how our design goals should best be realized. Additionally, the participants contributed

by providing feedback to the usability of our technical probe and by identifying design trade-offs with regards to our design goals.

Process. The first building block handed to the participants was a predesigned project in a familiar domain to make it easier for the participants to understand the design project. We implemented the design project with Figma [12]. Our participants were able to explore the design project. The second building block consisted of the two artifacts in the form of plugins we designed to support annotating design components and defining the tasks for the asynchronous usability tests. Both plugins were shown to the participants as static mock-ups. Participants were able to explore the artifacts and ask questions regarding functionality. The third building block shown to the participants was the artifact we designed to collect usage test data while evaluators tested the design prototype built in the prototyping tool. With this application, we collected real-world usage test data previous to our formative study to provide participants with realistic data. Our fourth building block shown was a prototype of the result plugin in the prototyping tool. Here we limited our-self to present descriptive data from the usability tests.

Participants. Using a student panel, we recruited four participants (three male, one female) up to this date. The participants self-reported an average age of 24.5 years (SD: 0.5), 1.1 years (SD: 1.1) of experience with visual prototyping design tasks, 0.9 years (SD: 1.2) of experience with prototyping design evaluation. Three participants were enrolled in graduate studies, one participant just finished graduate studies. We have deliberately chosen students as proxy for novice designers, as we believe that assistance is particularly important for this target group. Participants were paid approx. 16 USD in local currency (15 EUR) per hour of participation.

4.2 Results

In the following we describe key results from the formative study in the form of design trade-offs.

Order of actions in the proposed process. The first area where our participants suggested a trade-off relates to the simplicity of the process and the order of actions. The participants highlighted that the process is *"easy to understand for the designer [and test evaluator], as the steps are well defined", "easy and quick to implement"* and may *"lead to shorter development cycles [for inexperienced designers]"*. However, competing with simplicity, our participants suggested resolving the rigidity of the order of *Actions 1, 2, and 3*. In our initial version presented to the participants, *Actions 1, 2, and 3* were presented as sequential *steps*. Naturally, a design must already be created before the annotation can start. Nevertheless, even after annotating and task design, the *action design* can be performed before the user tests. Additionally, participants suggested allowing for tasks to be created before the annotation of components, so that designers can focus on annotating only components related to a previously constructed task. Furthermore, based on the feedback of the participants we decided to use the terminology *actions* instead of *steps*. This should highlight that actions do not have to be carried out sequentially. This is a design trade-off as simplicity for inexperienced prototype developers may eventually be reduced.

Integration of results into prototyping tools. Our participants reflected positively that no external tool may be needed to set up usability tests and to view the results. Again, most of the

positive arguments mentioned for the tight integration approach referred to simplicity. However, our participants expressed a trade-off between simplicity through integration in our process and the resulting complexity of designing user interfaces to present usability test data in a prototyping tool like Figma due to limited screen space. Therefore, data must be presented in a very compressed way. Our participants discussed possible solutions, e.g., to present only interpreted data first and only show raw data when needed. We see the need for further research to design interfaces for presenting usability test data in a way that preserves simplicity while being able to present all relevant usability test data linked to components in the prototyping tool.

Annotation of components. The last trade-off identified is that novice designers who can particularly benefit from actionable results may need help choosing the design components and annotating them correctly during *Actions 1 and 2*. Our participants discussed forms of enabling novice users to annotate components correctly. We iterated our prototype interface of *Action 2* and included an overview of multiple components using Google's *Material Design* [14] components. Nevertheless, future research should explore novel or existing approaches (e.g., [22]) that support (semi-) automated annotation in this context. Additionally, we learned that annotations can potentially present positive side benefits for novice users (e.g., support learning the existence of different components and reflecting on the components' correct use).

5 CONCLUSION AND FUTURE WORK

In recent years, research has introduced multiple assistance approaches and tools to help designers prototyping GUIs. However, research so far has been sparse on how to connect prototyping with quantitative asynchronous usability testing. There is a lack of research assisting designers in conducting, integrating into existing tools, and providing actionable support to use the results of usability tests for improving the prototype. In this paper, we distill three design goals to close this gap. We describe our approach that supports the three design goals. Specifically, we propose a process model and present several user interfaces. Additionally, we conducted a formative study to collect feedback for our approach. While our formative study provided insights into how assistance for quantitative test-driven prototyping could be realized, several aspects still need to be addressed in future research. Building on our results in the formative study, we plan to extend our results using a mixed-methods approach (such as in [33]) to further validate our design goals and evaluate the overall process (feature-based and against benchmarks). In our formative study, we identified interesting trade-offs that future research should address. Identified as one trade-off, leveraging quantitative descriptive data from usability tests within prototyping is challenging with regards to finding a good balance between providing comprehensive insights and overwhelming designers. Subsequently, another research opportunity is to understand how quantitative raw data from usability testing can be processed and interpreted to generate further actionable insights about potential usability improvements. We plan to carry out subsequent formative and summative studies to better understand and identify in which form test data should be integrated into the prototyping tool and linked to the components of the design.

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